
XData /sql

User Guide & Handbook

Automated SQL Query Grading Platform

A complete, screenshot-based guide for
Administrators, Instructors and Students

First start in 30 seconds: run `docker compose up --build`, open <http://localhost>,
and sign in with the seeded administrator account `admin1 / admin1`.

Contents

1	Introduction	2
2	Getting started	2
2.1	Starting the application	2
2.2	First-time initialisation	2
2.3	Signing in	2
3	Administrator guide	4
3.1	Creating users	4
3.2	Importing many users at once (CSV)	6
3.3	Managing existing users	6
3.4	Managing courses	8
4	Instructor guide	9
4.1	Defining schemas	10
4.2	Assignments and questions	11
4.3	Monitoring progress — the Gradebook	12
4.4	Other instructor tools	13
5	Student guide	16
5.1	The learning dashboard	16
5.2	Opening an assignment	17
5.3	Solving a question	18
5.4	Understanding the result	21
6	SQL Lab (Diagnostic Lab)	21
7	Dataset Playground	23
7.1	Mutation types	25
8	Tips & troubleshooting	26

1 Introduction

XData-Web is a web platform for learning SQL and grading SQL queries *automatically*. Instead of comparing query *text*, it checks whether a student's query is *semantically equivalent* to the instructor's reference solution, and — even when it is not — awards *partial credit* based on how close the query structure is (tables, columns, conditions, joins, grouping, ...).

This handbook walks through every part of the application, organised by the three user roles:

- **Administrator** — manages users and courses for the whole system.
- **Instructor** — creates courses, schemas, assignments and questions, and monitors student progress.
- **Student** — solves SQL questions and receives instant feedback.

Two additional tools are available to everyone who is signed in: the **SQL Lab** (a grading simulator) and the **Dataset Playground** (a test-data generator).

2 Getting started

2.1 Starting the application

The only prerequisite is **Docker**. From the project root, run:

```
docker compose up --build
```

This starts the database, the backend and the frontend. On the first start the database schema and a default administrator are created automatically, so there is no manual database setup. When the logs settle, open:

- **The app:** <http://localhost>
- **API docs (Swagger):** <http://localhost:8080/swagger-ui.html>

2.2 First-time initialisation

On a fresh installation exactly **one** account exists — the administrator. **Initialise the system by signing in with** `admin1 / admin1`.

Recommended first steps after the initial login:

1. Change the administrator password (key icon, top right).
2. Create a course (Admin Center → *Courses*).
3. Create at least one instructor and assign them to the course.
4. Let the instructor add a schema, an assignment and questions.
5. Create students and assign them to the course.

2.3 Signing in

Open the app and enter your user ID and password. New users receive their credentials from an administrator; the administrator starts with `admin1/admin1` (see §2.2).

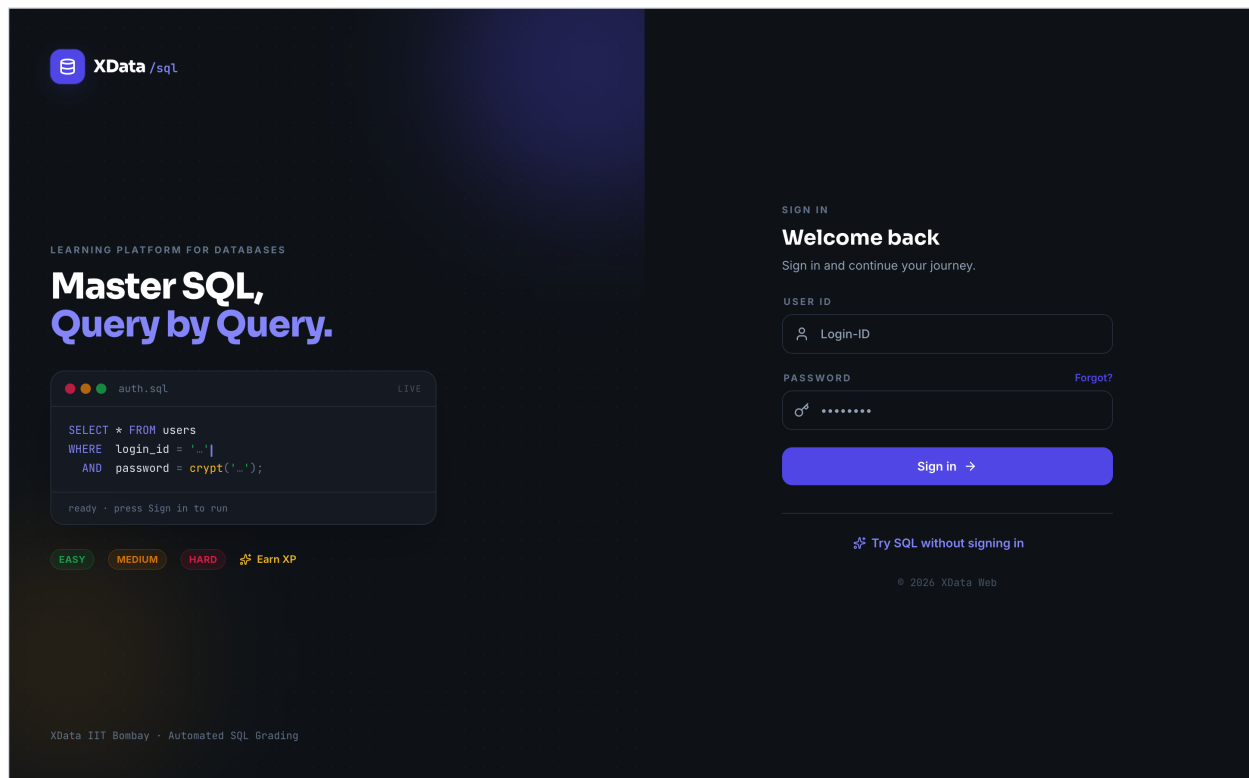


Figure 1: The login screen. Sign in with your user ID and password.

The link “*Try SQL without signing in*” opens a public SQL sandbox — handy for trying the editor without an account.

A freshly created account is asked to **set its own password on first sign-in** before continuing — a security default you cannot skip.

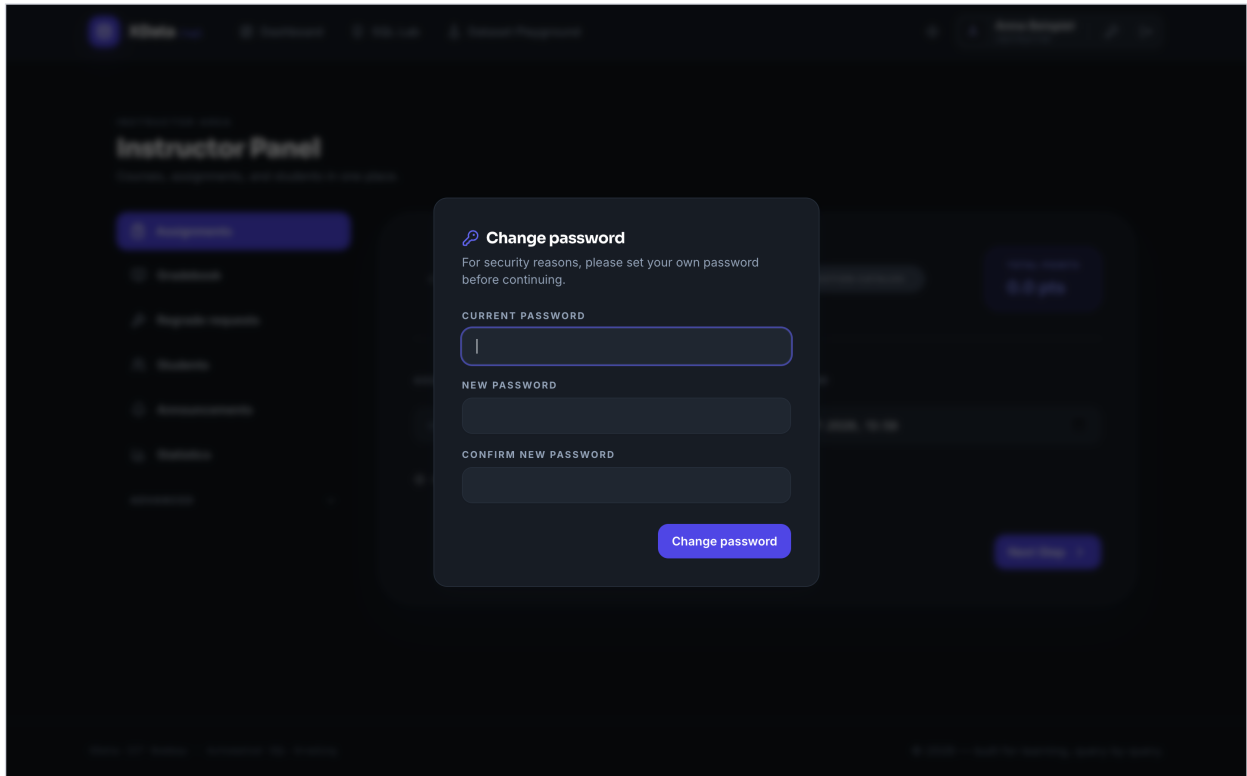


Figure 2: First sign-in: new users must choose their own password before they can use the app.

3 Administrator guide

After signing in as an administrator you land in the **Admin Center**, which has two tabs: *Users* and *Courses*.

3.1 Creating users

The *Users* tab contains the **Create new user** form on the left and the list of existing users on the right.

The screenshot displays the 'Admin Center' interface for user management. At the top, there's a navigation bar with 'XData /sql', 'Dashboard', 'SQL Lab', and 'Dataset Playground'. The 'Admin Center' section is titled 'System-wide management & monitoring.' and includes tabs for 'Users' and 'Courses'. Below this, the 'User management' section features buttons for 'Template' and 'Import CSV'.

The 'Create new user' form is prominently displayed. It includes fields for 'FULL NAME' (e.g. John Doe), 'LOGIN ID' (e.g. jdoe), 'EMAIL ADDRESS' (john@example.com), 'PASSWORD' (masked with dots), and 'ROLE' (STUDENT). There's also a 'COURSE ASSIGNMENT' section with a dropdown menu showing 'Datenbanken I'. A 'Create user' button is at the bottom right of the form.

Below the form is a table titled 'Assigned users' with 5 total users. The table has columns for STATUS, NAME, LOGIN ID, COURSE, ROLE, CREATED, and ACTIONS. The users listed are:

STATUS	NAME	LOGIN ID	COURSE	ROLE	CREATED	ACTIONS
ACTIVE	S schülerr	schülerr	DB1-2026	STUDENT	6/21/2026	[Icons for edit, delete, etc.]
ACTIVE	E Erika Lehrer	Lehrer1	DB1-2026	INSTRUCTOR	6/20/2026	[Icons for edit, delete, etc.]
ACTIVE	M Max Schüler	schueler1	DB1-2026	STUDENT	6/20/2026	[Icons for edit, delete, etc.]
ACTIVE	D Daniel Instructor	daniel	No course	INSTRUCTOR	6/17/2026	[Icons for edit, delete, etc.]
ACTIVE	A Admin	admin1	No course	ADMIN	6/17/2026	[Icons for edit, delete, etc.]

The footer of the page indicates 'XData IIT Bombay · Automated SQL Grading' and '© 2026 — built for learning, query by query.'

Figure 3: Admin Center → Users: the user-creation form and the user list with per-row actions.

To create a single user:

1. Fill in **Full name**, **Login ID** (used to sign in) and **Email**.
2. Set an initial **Password**.
3. Choose a **Role**: STUDENT, INSTRUCTOR or ADMIN.
4. Tick one or more courses under **Course assignment**.
5. Click **Create user**. The new account appears in the list immediately.

The screenshot shows the 'Admin Center' interface for user management. The 'Create new user' form is the central focus, with fields for Full Name, Login ID, Email Address, Password, Role, and Course Assignment. The 'Assigned users' table below shows a list of users with columns for Status, Name, Login ID, Course, Role, Created, and Actions.

STATUS	NAME	LOGIN ID	COURSE	ROLE	CREATED	ACTIONS
ACTIVE	Anna Beispiel	anna1	Datenbanken I	INSTRUCTOR	2024-10-26	[Icons for actions]

Figure 4: The create-user form filled in for a new instructor, with a course ticked.

Roles in short: STUDENTS solve questions; INSTRUCTORS manage courses, schemas, assignments and grading; ADMINS manage all users and courses.

3.2 Importing many users at once (CSV)

For whole classes, use **Import CSV** (top right of the Users tab). Click **Template** first to download a correctly-formatted CSV, fill in one row per user, then upload it with **Import CSV**. All users are created in one step.

3.3 Managing existing users

Each row in the user list offers the following actions:

- **Disable / Enable** — block or restore sign-in without deleting the account.
- **Sign in as this user** — *impersonate* the user to see exactly what they see (great for support and for verifying an instructor's setup). A confirmation dialog appears first.
- **Edit** — change name, email, role or course assignment.
- **Reset password** — set a new password for the user.
- **Delete** — permanently remove the account.

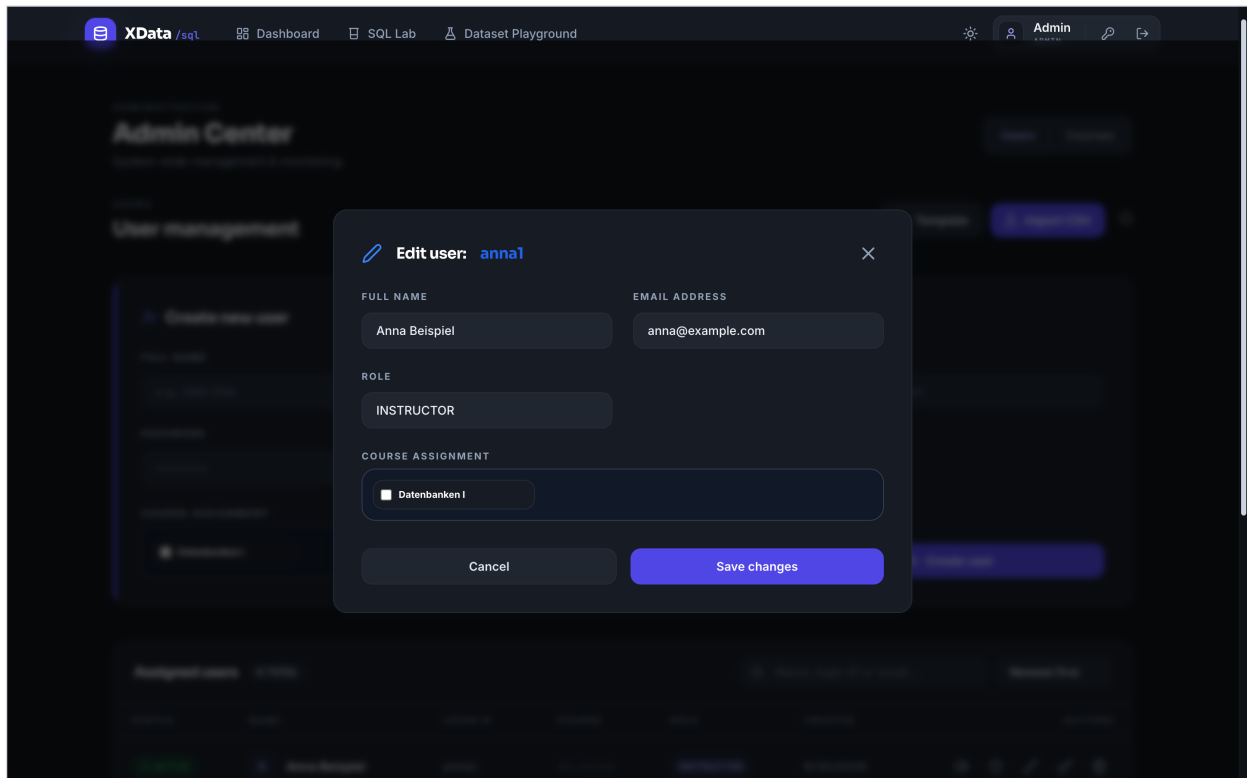


Figure 5: The Edit-user dialog: change name, email, role and course assignment.

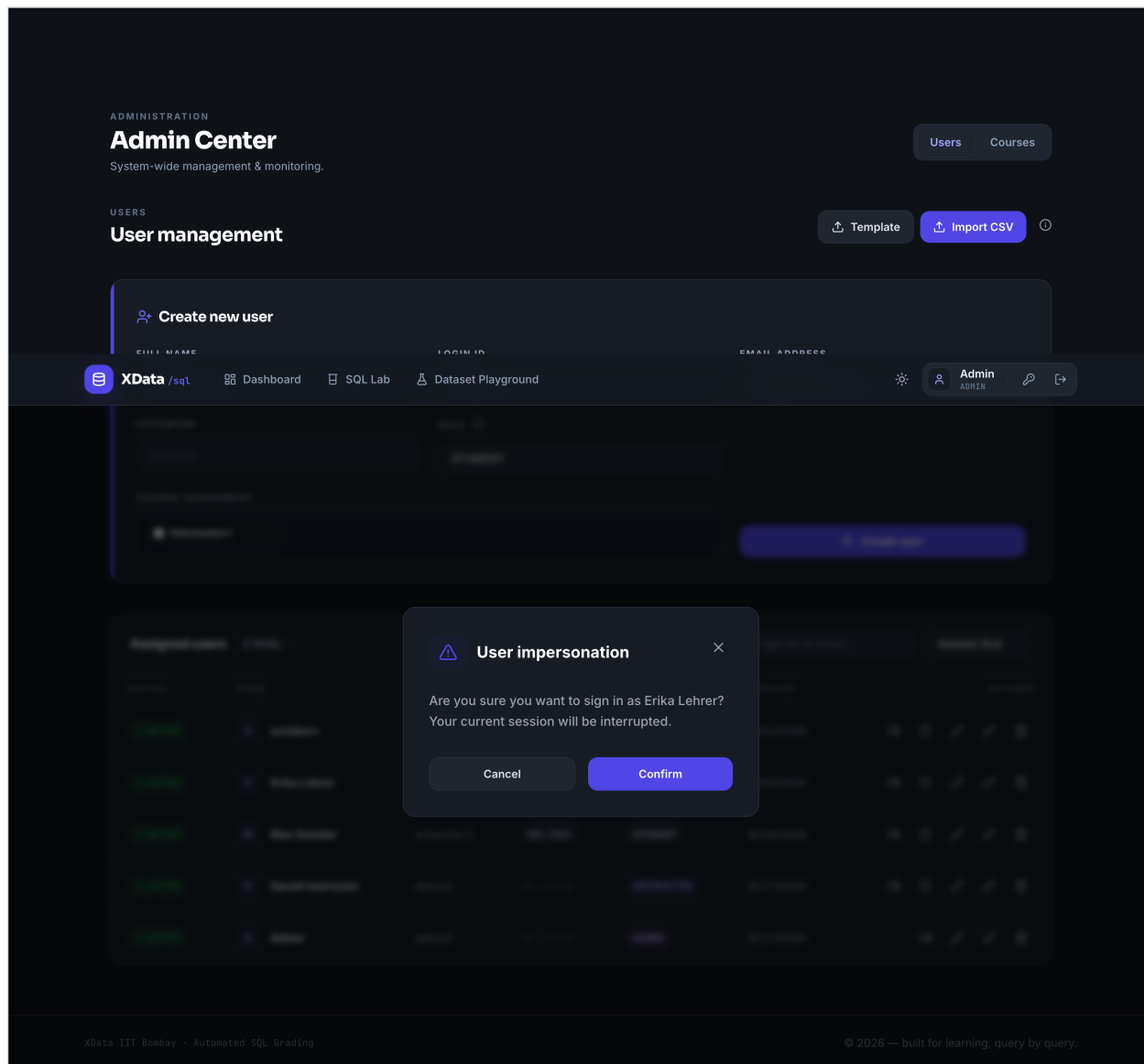


Figure 6: “Sign in as this user” asks for confirmation before impersonating — your admin session is replaced by the target user’s view.

3.4 Managing courses

The *Courses* tab lists all courses and lets you create new ones. A course bundles its instructors, students, schemas and assignments.

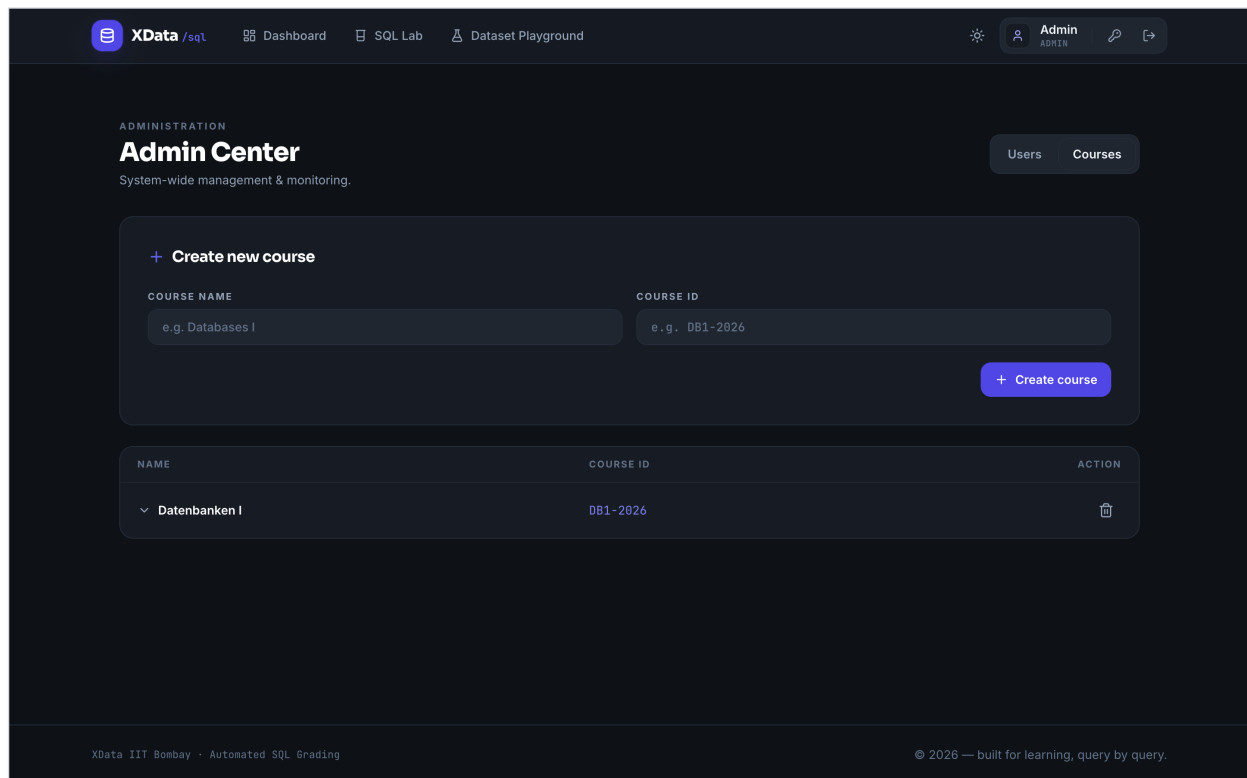


Figure 7: Admin Center → Courses: create and manage courses.

4 Instructor guide

Instructors work in the **Instructor Panel**. The left sidebar groups every tool; the *Advanced* section holds the setup tools (course materials, SQL schemas, databases, courses).

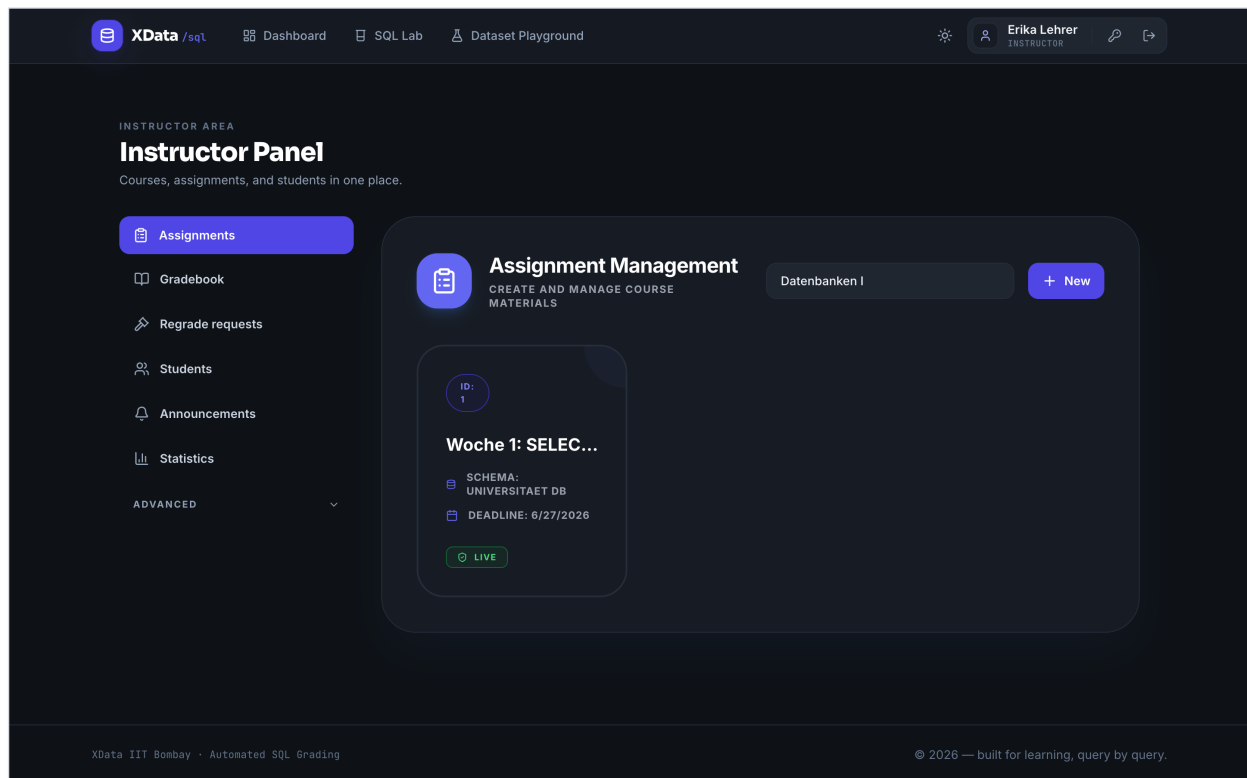


Figure 8: The Instructor Panel. The sidebar gives access to assignments, gradebook, students, statistics and the advanced setup tools.

4.1 Defining schemas

Before creating questions, define the database schema the questions run against. Open *Advanced* → *SQL Schemas*, select the course, and add the schema (table definitions). Schemas are reused by assignments, the SQL Lab and the Dataset Playground.

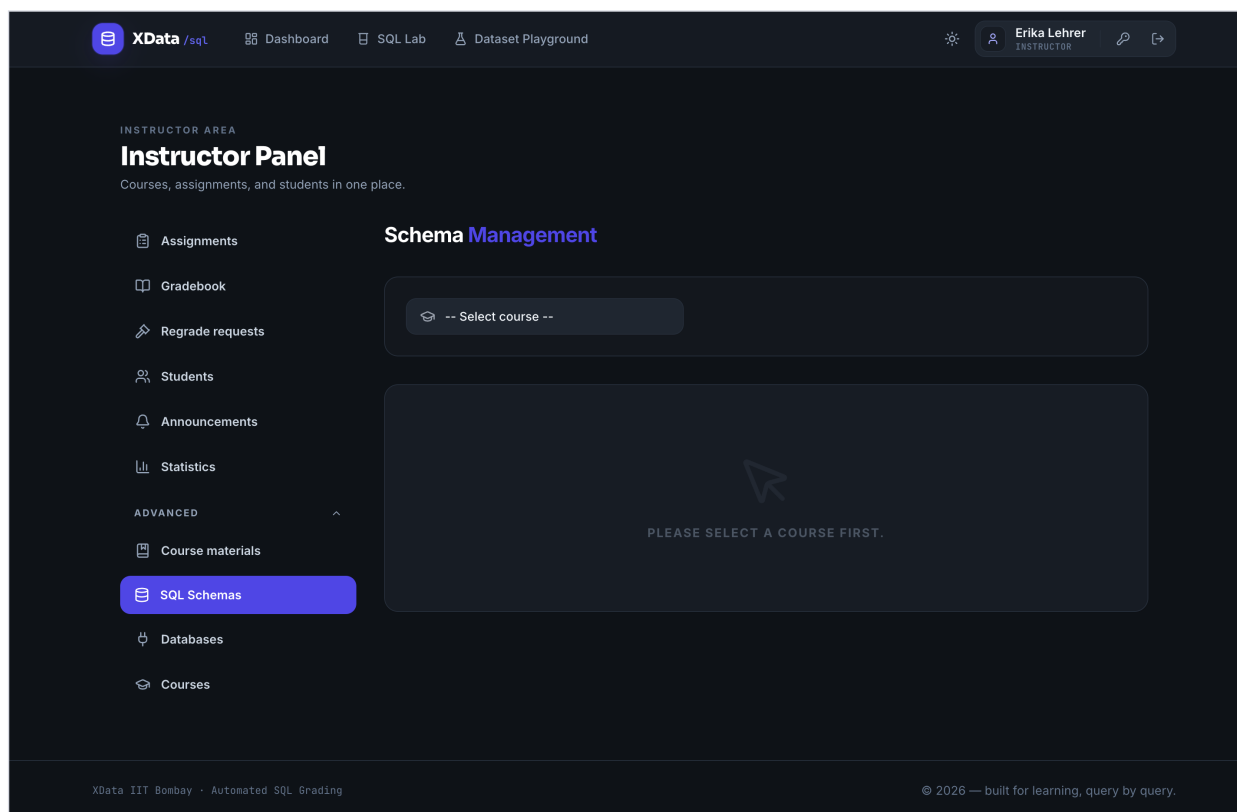


Figure 9: Schema management. Select a course, then create or edit the SQL schema used by its questions.

4.2 Assignments and questions

Open *Assignments*, pick the course, and click **New** to create an assignment. For each assignment you set a **schema** and a **deadline**, and add **questions**. Every question has a problem statement, a reference (model) solution, a difficulty, point value and optional hints. From an assignment you can also **Edit**, **Duplicate**, view **Statistics**, grant **Deadline extensions**, or **Delete** it. Set an assignment to *LIVE* to make it visible to students.

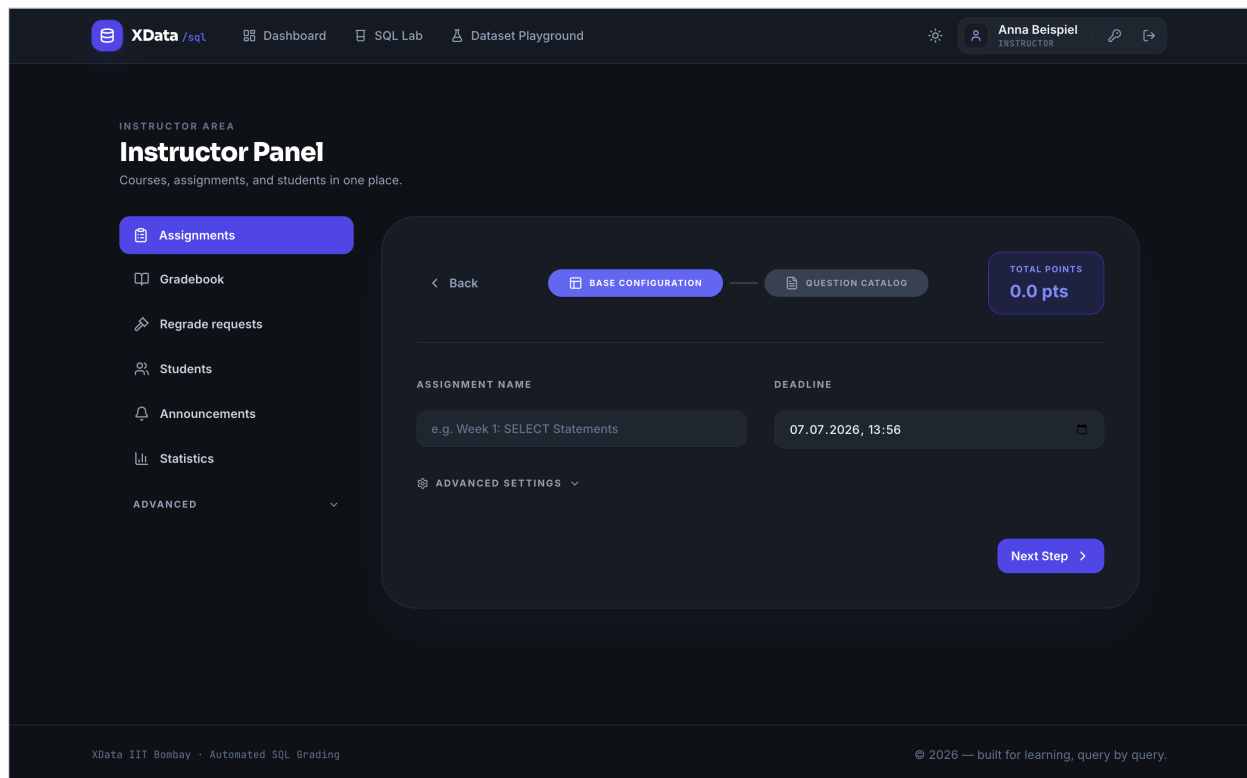


Figure 10: Creating an assignment: a two-step wizard — base configuration (name, deadline) then the question catalogue.

4.3 Monitoring progress — the Gradebook

The *Gradebook* shows every student's score per assignment, plus a total, and can be exported as **CSV**.

INSTRUCTOR AREA
Instructor Panel
 Courses, assignments, and students in one place.

GRADING
Gradebook Datenbanken I (DB1-2026) ↓ CSV

STUDENT	WOCHE 1: SELECT BASICS	TOTAL
Max Schüler	10/10	100%
schülerrrr	0/10	0%

ADVANCED

- Course materials
- SQL Schemas
- Databases
- Courses

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Figure 11: The Gradebook: per-student, per-assignment scores with a CSV export.

4.4 Other instructor tools

- **Regrade requests** — review and resolve student requests to re-check a submission.
- **Students** — see the roster of the selected course.
- **Announcements** — post messages to a course.
- **Statistics** — per-assignment analytics (open them via the chart icon on an assignment).
- **Databases** — register external database connections used for grading.
- **Course materials** — attach links and resources to a course.

The screenshot displays the 'Instructor Panel' in a dark-themed web application. The top navigation bar includes the 'XData /sql' logo, a 'Dashboard' button, and links to 'SQL Lab' and 'Dataset Playground'. The user 'Anna Beispiel' is logged in as an 'INSTRUCTOR'. The main section is titled 'INSTRUCTOR AREA' and 'Instructor Panel', with a subtitle 'Courses, assignments, and students in one place.' A left sidebar contains navigation links: 'Assignments', 'Gradebook', 'Regrade requests', 'Students' (highlighted), 'Announcements', and 'Statistics'. The main content area is 'COURSE PARTICIPANTS' and 'User management', featuring buttons for 'Template', 'Add existing', and 'Import CSV'. A 'Create new student' form is visible, with fields for 'FULL NAME', 'LOGIN ID', 'EMAIL ADDRESS', 'PASSWORD', and 'COURSE ASSIGNMENT (MY COURSES)'. Below the form is a table of 'Assigned users' for the course 'DB1-2026', showing 4 total users. The table columns are STATUS, NAME, LOGIN ID, COURSE, ROLE, CREATED, and ACTION. The users listed are Anna Beispiel (Instructor), schülerrr (Student), Erika Lehrer (Instructor), and Max Schüler (Student).

INSTRUCTOR AREA

Instructor Panel

Courses, assignments, and students in one place.

Assignments

Gradebook

Regrade requests

Students

Announcements

Statistics

ADVANCED

COURSE PARTICIPANTS

User management

[Template](#) [Add existing](#) [Import CSV](#)

Create new student

FULL NAME: e.g. John Doe LOGIN ID: e.g. jdoe EMAIL ADDRESS: john@example.com

PASSWORD: ***** COURSE ASSIGNMENT (MY COURSES): ☒ Datenbanken I [Create student](#)

Assigned users 4 TOTAL

Q Name, login ID or email... [Newest first](#)

STATUS	NAME	LOGIN ID	COURSE	ROLE	CREATED	ACTION
ACTIVE	A Anna Beispiel	anna1	DB1-2026	INSTRUCTOR	6/30/2026	View Edit Delete
ACTIVE	S schülerrr	schülerrr	DB1-2026	STUDENT	6/21/2026	View Edit Delete
ACTIVE	E Erika Lehrer	lehrer1	DB1-2026	INSTRUCTOR	6/20/2026	View Edit Delete
ACTIVE	M Max Schüler	schueler1	DB1-2026	STUDENT	6/20/2026	View Edit Delete

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Figure 12: Students: the roster of the selected course.

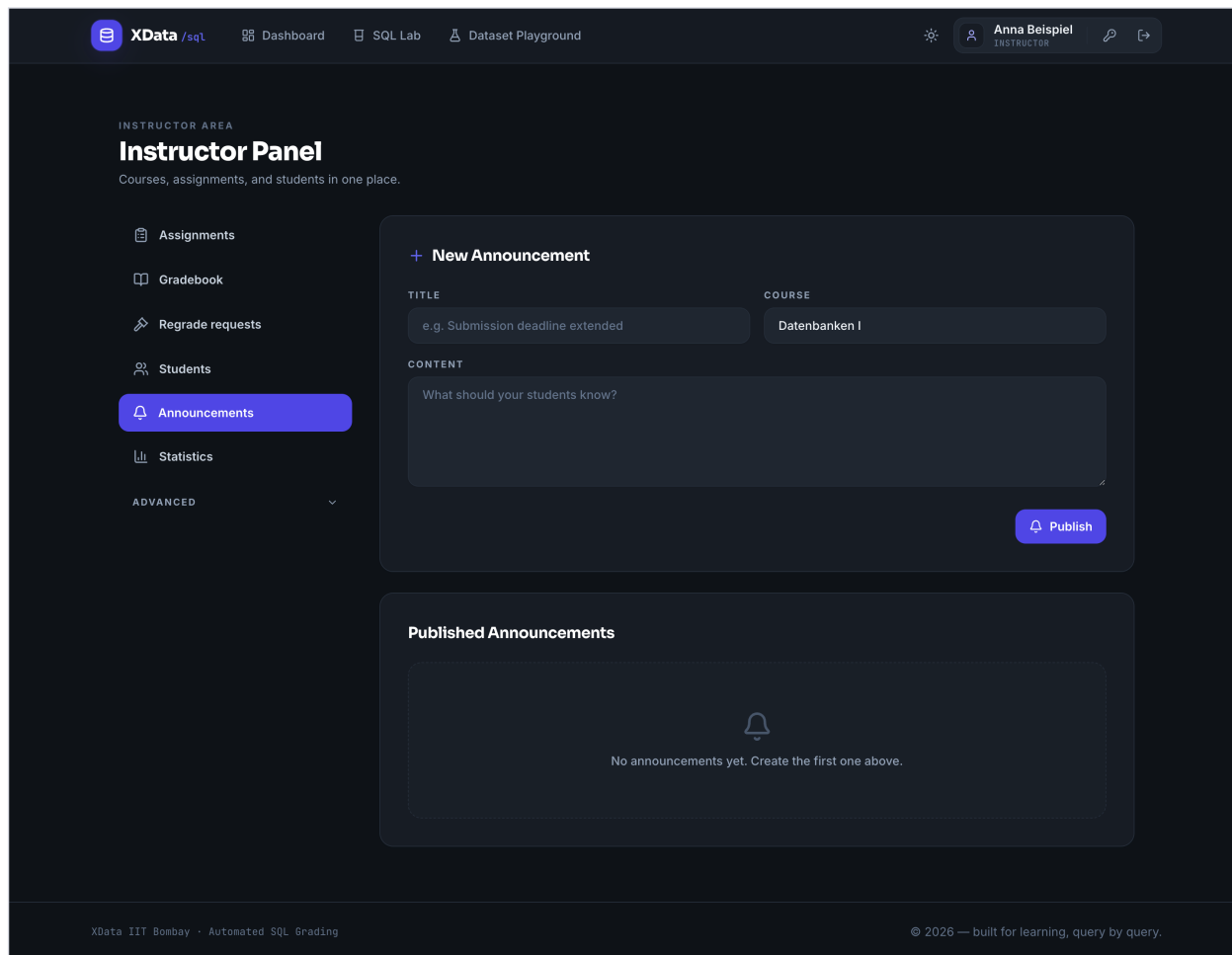


Figure 13: Announcements: post a message (title, course, content) to a course; published announcements are listed below.

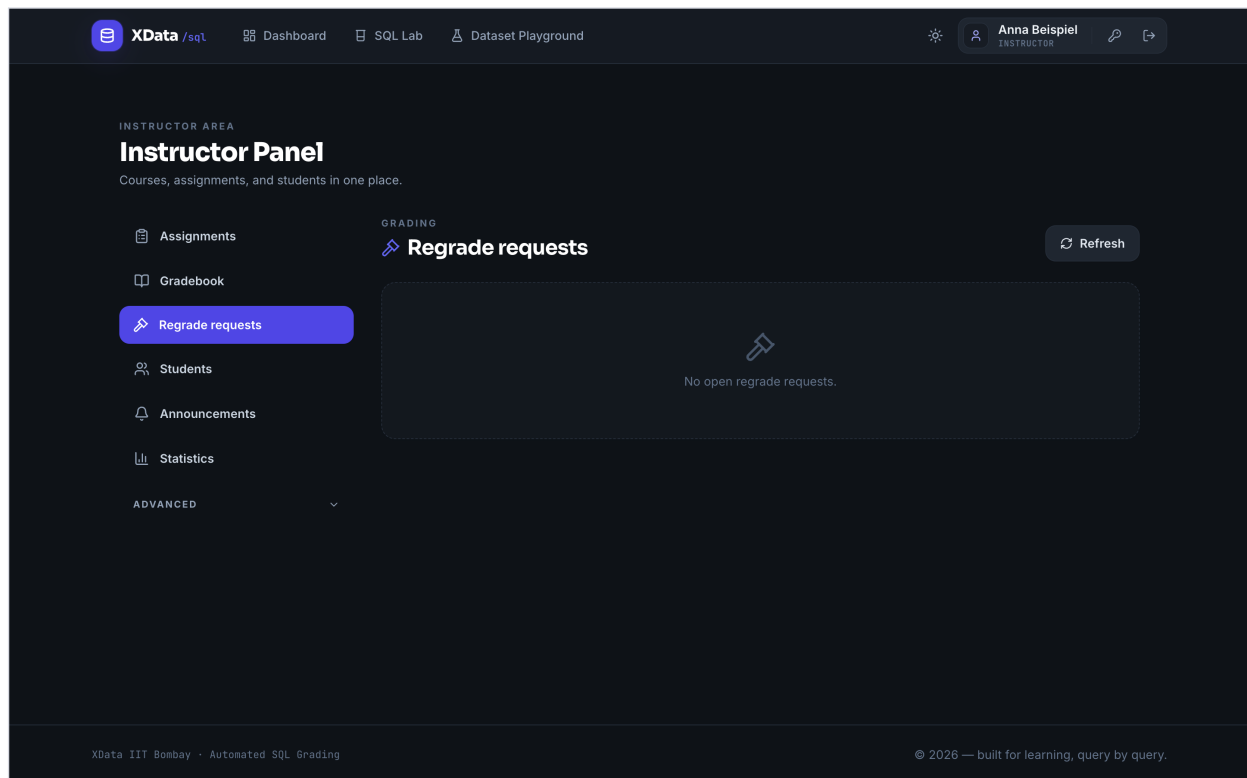


Figure 14: Regrade requests: review and resolve students' requests to re-check a submission.

5 Student guide

5.1 The learning dashboard

After signing in, a student sees their personal dashboard: level and XP progress, a daily streak, a **leaderboard**, earned **badges**, and the list of **assignments**.

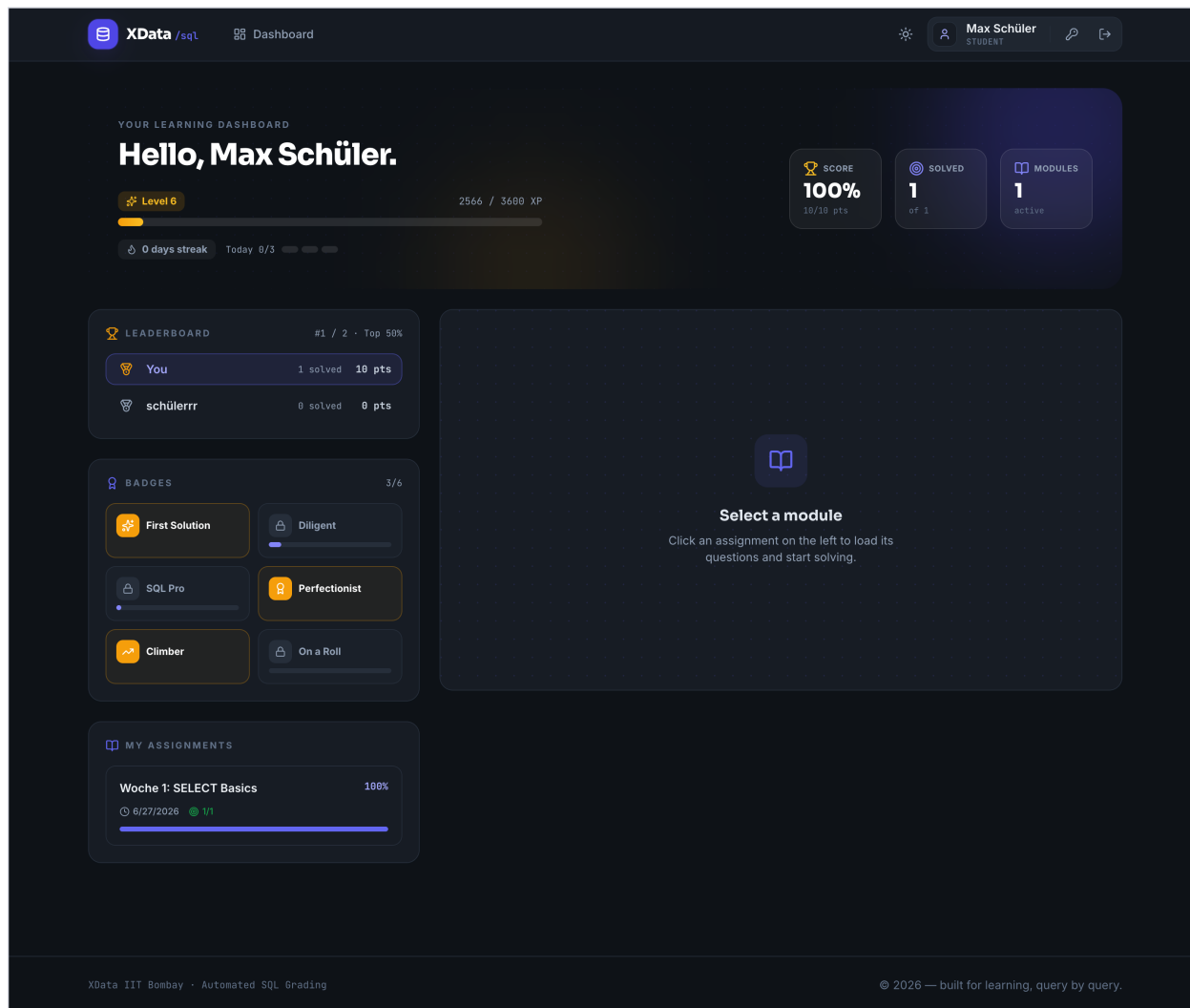


Figure 15: The student dashboard: XP and level, leaderboard, badges and assigned modules.

5.2 Opening an assignment

Click an assignment under *My assignments* to load its questions. Each question shows its difficulty, tags and point value; solved questions are marked with a green check.

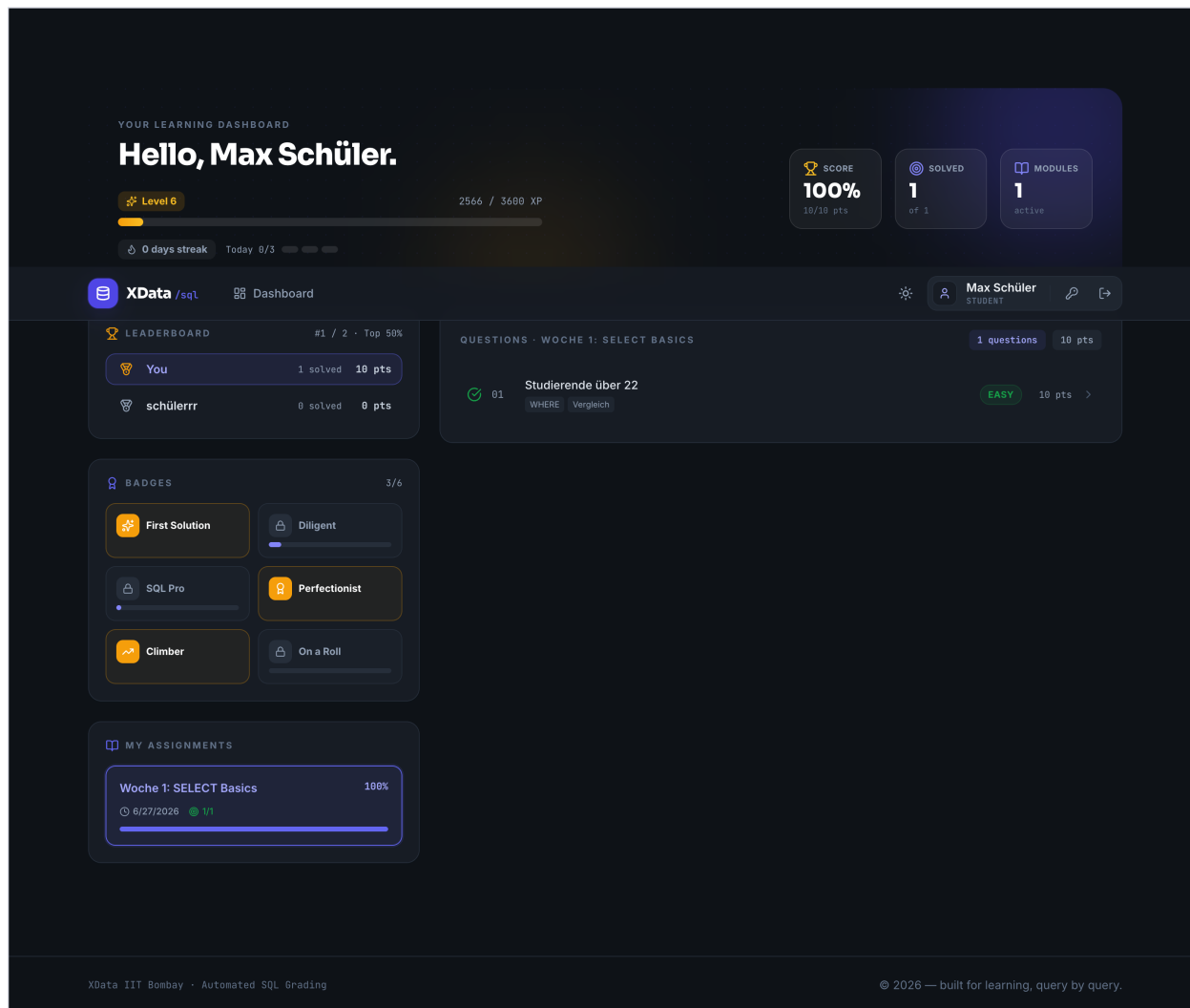


Figure 16: Selecting an assignment loads its question list on the right.

5.3 Solving a question

Open a question to reach the solving view. It contains:

- The **problem statement**, plus **Schema**, **Cheat sheet** and **Sample data** helpers.
- Step-by-step **hints** (optional — revealing them may cost points).
- The **SQL editor** with syntax highlighting and auto-completion.
- **Run** (try the query) and **Submit answer & check** (grade it).

The screenshot displays the XData-Web student interface. At the top, the user is greeted as 'Max Schüler' with a Level 6 badge and a progress bar showing 2544 / 3400 XP. The dashboard includes a 'LEADERBOARD' with 'You' at the top (1 solved, 10 pts) and 'schülerr' below (0 solved, 0 pts). A 'BADGES' section shows 'First Solution', 'Diligent', 'SQL Pro', 'Perfectionist', 'Climber', and 'On a Roll'. The 'MY ASSIGNMENTS' section shows 'Woche 1: SELECT Basics' at 100% completion.

The main content area is titled 'QUESTIONS · WOCHE 1: SELECT BASICS' and shows a question 'Studierende über 22' (EASY, 10 pts). The problem statement is: 'Gib die Namen aller Studierenden aus, die älter als 22 Jahre sind.' The hints section says 'Stuck? Reveal hints step by step.' and 'Show hint (0/2)'. The SQL editor shows a single line '1'. The 'Run' button is disabled, and the 'Submit answer & check' button is active.

The 'LAST RESULT' section shows a '100%' score and an 'OVERALL SCORE' of '3.0 / 3.0'. Below this is a table comparing the user's solution to the model solution:

COMPONENT	YOUR SOLUTION	MODEL SOLUTION
Tables (FROM)	✓ STUDENTS (+1.0)	STUDENTS
Columns (SELECT)	✓ name (+1.0)	name
Conditions (WHERE)	✓ age > 22 (+1.0)	age > 22
Joins (JOIN)	-	-
Grouping (GROUP BY)	-	-
Sorting (ORDER BY)	-	-
Filter (HAVING)	-	-
Duplicates (DISTINCT)	✓ No (-0.0)	NO

At the bottom of the result section are buttons for 'Compare expected output' and 'Request regrade'.

Figure 17: The question view: problem statement, hints, SQL editor, and the structural result with per-component feedback.

Helpers. The **Schema** button reveals the tables, columns, primary keys and relationships of the question's database — without leaving the page. **Sample data** and **Cheat sheet** give example rows and a quick SQL reference.

YOUR LEARNING DASHBOARD

Hello, Max Schüler.

Level 6 2566 / 3488 XP

Score 100% Solved 1 Modules 1

5 days streak Today 8/1 100 100 100

LEADERBOARD #1 / 2 · Top 50%

You 1 solved 10 pts

schülerrr 0 solved 0 pts

BADGES 3/6

First Solution Diligent

SQL Pro Perfectionist

Climber On a Roll

MY ASSIGNMENTS

Woche 1: SELECT Basics 100%

6/27/2028 100

QUESTIONS · WOCHE 1: SELECT BASICS 1 questions 10 pts

Studierende über 22 100% 10 pts

Problem Statement

Gib die Namen aller Studierenden aus, die älter als 22 Jahre sind.

HINTS

Stuck? Reveal hints step by step. Show hint (0/2)

SCHEMA: UNIVERSITY DB

STUDENTS 4 columns

id	name	age	major	gpa
INTEGER	VARCHAR(100)	INTEGER	VARCHAR(100)	NUMERIC(3,1)

COURSES 3 columns

course_id	title	credits
INTEGER	VARCHAR(100)	INTEGER

ENROLLMENTS 4 columns

st_	course_id	grade
INTEGER	INTEGER	NUMERIC(3,1)

Primary key Foreign key (relationship)

1

Run Submit answer & check

LAST RESULT 100%

Overall score 3.0 / 3.0 100%

MAIN QUERY

COMPONENT	YOUR SOLUTION	MODEL SOLUTION
Tables (FROM)	✓ STUDENTS (1/1)	STUDENTS
Columns (SELECT)	✓ name (1/1)	name
Conditions (WHERE)	✓ age > 22 (1/1)	age > 22
Joins (JOIN)	-	-
Grouping (GROUP BY)	-	-
Sorting (ORDER BY)	-	-
Filter (HAVING)	-	-
Duplicates (DISTINCT)	✓ No (0/2)	NO

Compare expected output Request upgrade

XData 117 Berlin · PostgreSQL 10, Oracle

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Figure 18: The *Schema* helper expands the database structure (tables, columns, keys and foreign-key relationships) inline.

Hints. If you are stuck, reveal hints step by step with **Show hint**. Hints are optional and are unlocked one at a time.

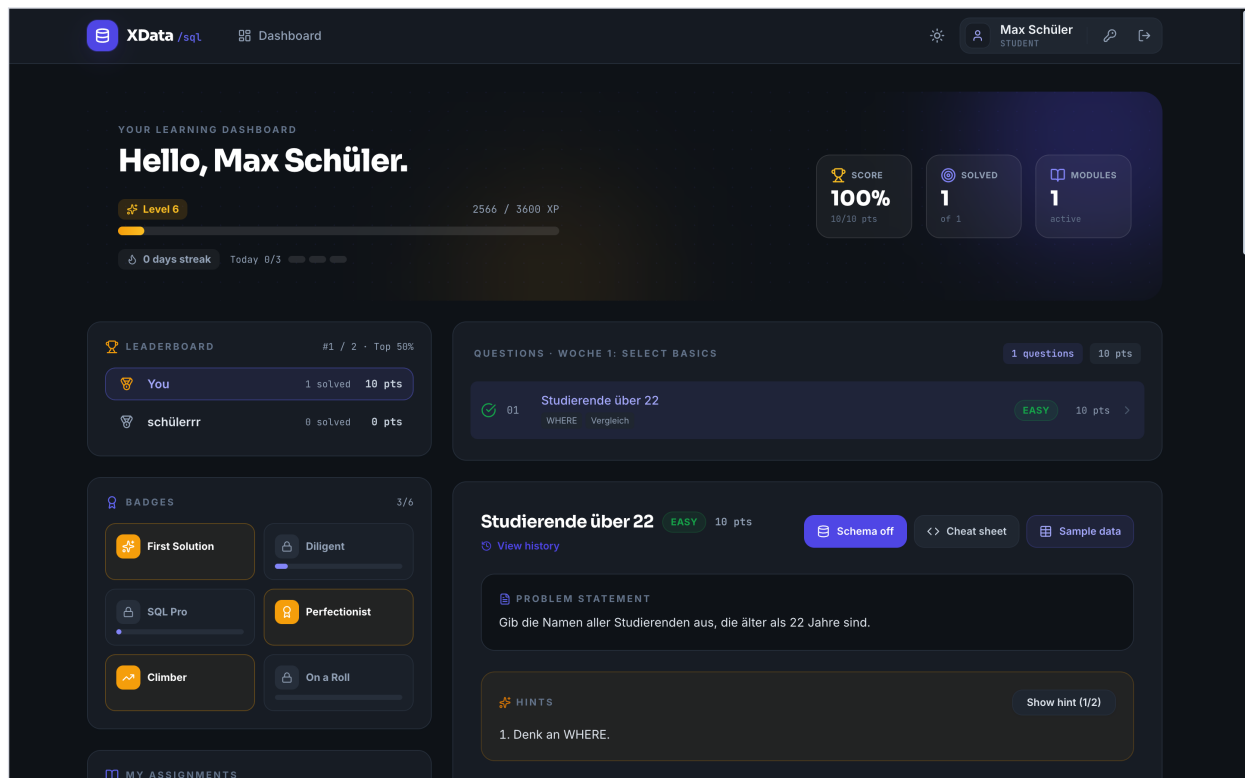


Figure 19: Hints are revealed one step at a time.

5.4 Understanding the result

After submitting, the **Last result** panel shows an overall score and a **component-by-component** comparison of your query against the model solution — tables, columns, conditions, joins, grouping, sorting, filters and duplicates. Each matching part earns partial credit, so a near-miss still scores. You can **Compare expected output** or, if you believe the grade is wrong, **Request a regrade** (the instructor reviews it under *Regrade requests*).

6 SQL Lab (Diagnostic Lab)

The **SQL Lab** is a grading *simulator*. Pick a schema, enter a model solution and a candidate query, optionally adjust the grading weights, and run a structural evaluation — the same engine that grades student submissions. It is ideal for instructors testing a question or for anyone exploring how the partial marking works.

The screenshot displays the SQL Lab (Diagnostic Lab) interface. At the top, there is a navigation bar with links to XData /sql, Dashboard, SQL Lab, and Dataset Playground. The user is logged in as Admin. The main heading is "SQL Diagnostic Lab" with a subtitle "Structural grading simulation for SQL solutions." and a student ID "students_1781727484". Below this is a section for "Adjust grading weights". The interface is split into two panels: "Model solution" (Instructor View) and "Student query" (Test Candidate). The Model solution panel shows a SQL query:

```
1 SELECT user_name FROM xdata_users WHERE internal_user_id = 'admin-uuid-admin1';
```

 The Student query panel shows a SQL query:

```
1 SELECT user_name FROM xdata_users;
```

 At the bottom, there is a large blue button labeled "Run structural evaluation". The footer contains the text "XData IIT Bombay · Automated SQL Grading" and "© 2026 — built for learning, query by query."

Figure 20: SQL Lab: a model solution (left) and a student/candidate query (right).

The screenshot displays the SQL Diagnostic Lab interface. At the top, there's a navigation bar with 'XData /sql', 'Dashboard', 'SQL Lab', and 'Dataset Playground'. The user is logged in as 'Admin'. The main heading is 'SQL Diagnostic Lab' with a subtitle 'Structural grading simulation for SQL solutions.' and a student ID 'students_1781727484'. Below this is a section to 'Adjust grading weights'. The interface is split into two panels: 'Model solution' (Instructor View) and 'Student query' (Test Candidate). The 'Model solution' shows a query: `1 SELECT user_name FROM xdata_users WHERE internal_user_id = 'admin-uuid-admin1';`. The 'Student query' shows a query: `1 SELECT user_name FROM xdata_users;`. A large blue button labeled 'Run structural evaluation' is positioned below the queries. The 'STRUCTURAL ANALYSIS & FEEDBACK' section shows an 'OVERALL SCORE' of 2.0 / 3.0 (67%). Below this is a table comparing the 'MAIN QUERY' components between 'YOUR SOLUTION' and 'MODEL SOLUTION'.

COMPONENT	YOUR SOLUTION	MODEL SOLUTION
Tables (FROM)	✓ XDATA_USERS (+1.0)	XDATA_USERS
Columns (SELECT)	✓ user_name (+1.0)	user_name
Conditions (WHERE)	-	internal_user_id = 'admin-uuid-admin1'
Joins (JOIN)	-	-
Grouping (GROUP BY)	-	-
Sorting (ORDER BY)	-	-
Filter (HAVING)	-	-
Duplicates (DISTINCT)	✓ No +0.0	NO

At the bottom, there's a footer with 'XData IIT Bombay · Automated SQL Grading' and '© 2026 — built for learning, query by query.'

Figure 21: SQL Lab result: an overall score plus a per-component structural breakdown.

7 Dataset Playground

The **Dataset Playground** generates targeted test datasets that expose errors (“mutants”) in a SQL query, using the SMT solver (Z3). Select a schema, paste a reference query, choose which **mutation types** to target, and click **Generate dataset**. The generated INSERT statements appear on the right and can be copied or downloaded as .sql.

Dataset Playground

Generate targeted datasets to detect mutants of your SQL query.

Load schema (.sql)

SELECT DATABASE SCHEMA

students_1781727484

REFERENCE QUERY (SQL)

1 e.g. SELECT name FROM students WHERE age > 20;

MUTATION TYPES

☒ Selection ☒ Joins (Equi) ☐ Joins (Non-Equi)

☒ Aggregation ☐ Distinct ☐ Group By

☐ Having

Generate dataset

GENERATED DATASET (SQL INSERTS)

No dataset generated yet.
Click "Generate dataset".

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© 2026 — built for learning, query by query.

Figure 22: Dataset Playground: choose a schema and mutation types, then generate a discriminating dataset.

Important — qualify your columns. The reference query must use *table-qualified* column names, e.g. `SELECT employees.name FROM employees WHERE employees.salary > 50000` (not just `name/salary`). This is a requirement of the underlying data-generation engine.

A successful run produces a dataset that includes *boundary* values — exactly the rows needed to tell a correct query apart from a subtly wrong one. For `employees.salary > 50000`, the engine generates rows with `salary = 50001` (passes), `50000` (the boundary that kills a `>=` mutant) and a non-matching value:

The screenshot shows the XData Dataset Playground interface. The top navigation bar includes links to XData /sql, Dashboard, SQL Lab, and Dataset Playground. The main header displays the 'Dataset Playground' title and a 'Load schema (.sql)' button. Below the header, the interface is divided into two main panels.

Left Panel:

- SELECT DATABASE SCHEMA:** A dropdown menu showing 'Company DB'.
- REFERENCE QUERY (SQL):** A text area containing the query: `1 SELECT employees.name FROM employees WHERE employees.salary >`.
- MUTATION TYPES:** A section with checkboxes for various mutation types:
 - ☒ Selection
 - ☒ Joins (Equi)
 - ☐ Joins (Non-Equi)
 - ☒ Aggregation
 - ☐ Distinct
 - ☐ Group By
 - ☐ Having
- Generate dataset:** A blue button with a play icon.

Right Panel:

- GENERATED DATASET (SQL INSERTS):** A text area showing the generated SQL inserts:


```
insert into DEPARTMENTS values ('3',null,'4')
insert into DEPARTMENTS values ('5',null,'6')
insert into EMPLOYEES values ('7',null,'8','50001',null)
insert into DEPARTMENTS values ('1',null,'21240')
insert into DEPARTMENTS values ('-92279',null,'3')
insert into EMPLOYEES values ('4',null,'5','-97560',null)
insert into DEPARTMENTS values ('3',null,'4')
insert into DEPARTMENTS values ('5',null,'6')
insert into EMPLOYEES values ('7',null,'8','50000',null)
```
- 9 test data row(s) generated.** A green status message at the bottom of the panel.

The footer of the interface includes 'XData IIT Bombay · Automated SQL Grading' on the left and '© 2026 — built for learning, query by query.' on the right.

Figure 23: A successful generation on a custom schema. Z3 produced boundary values (50000 / 50001) that discriminate the query from its mutants.

7.1 Mutation types

Each ticked **mutation type** adds one targeted “killing” dataset aimed at a specific class of mistake — *Selection* (WHERE), *Joins (equi / non-equi)*, *Aggregation*, *Distinct*, *Group By* and *Having*. Tick only the types relevant to your query: a type that does not apply (e.g. *Aggregation* for a query without a GROUP BY) simply adds no rows. Toggling a type changes the output — for the salary query above, enabling *Selection* adds the boundary-killing rows (9 rows total), while disabling it falls back to the base dataset (3 rows).

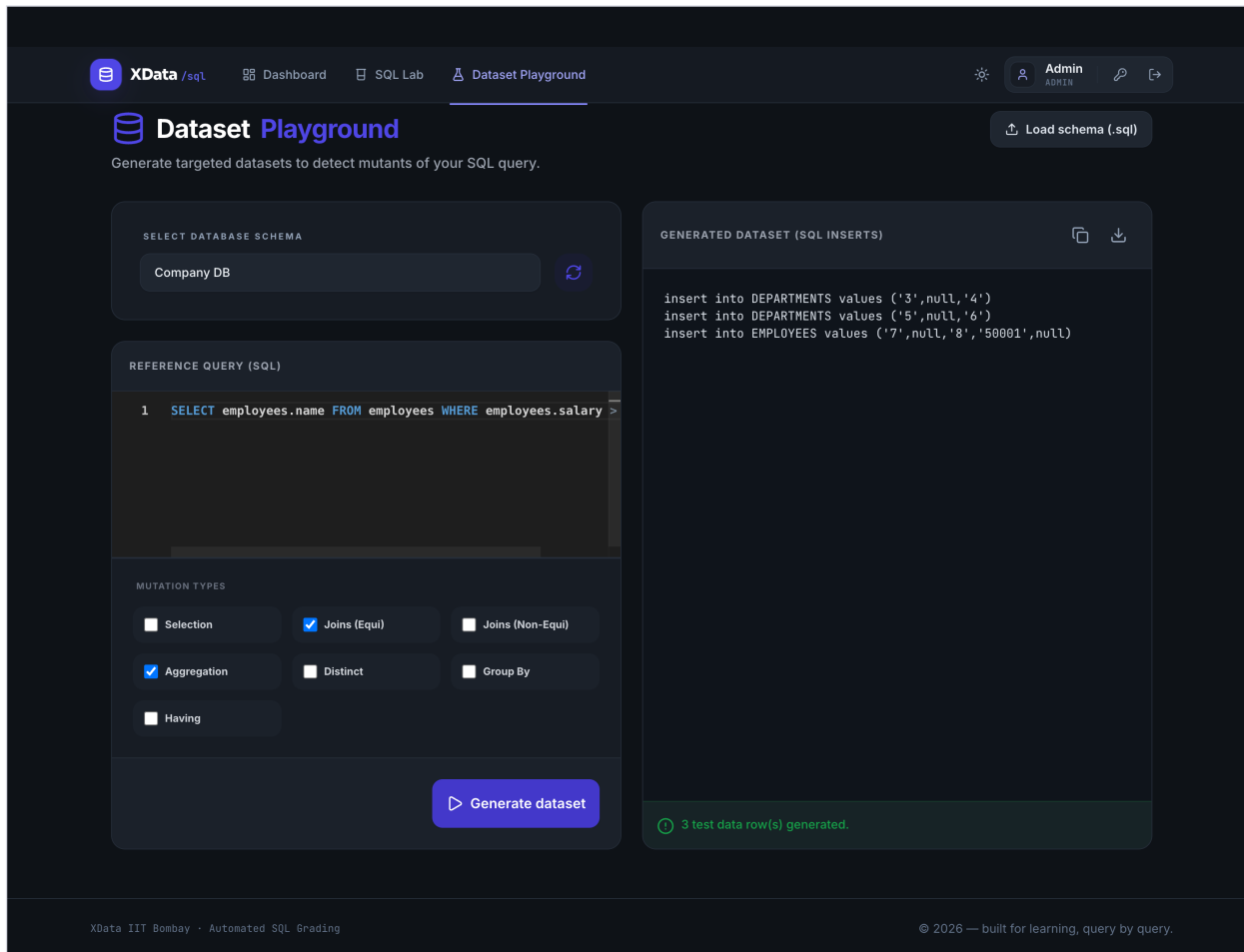


Figure 24: Mutation types are applied live: unticking a type removes its targeted rows from the generated dataset.

8 Tips & troubleshooting

- **Light / dark mode** — toggle with the sun/moon icon in the top bar (Figure 25).
- **Change password** — the key icon in the top bar opens the change-password dialog (Figure 26).
- **“No assignment / course selected”** — some panels (Statistics, Schema management) need you to pick a course or assignment first; this is expected.
- **Ports** — the app uses 80 (frontend), 8080 (backend) and 5433 (database). Make sure they are free.
- **Fresh start** — `docker compose down -v` wipes the database and re-seeds the `admin1` account on the next start.

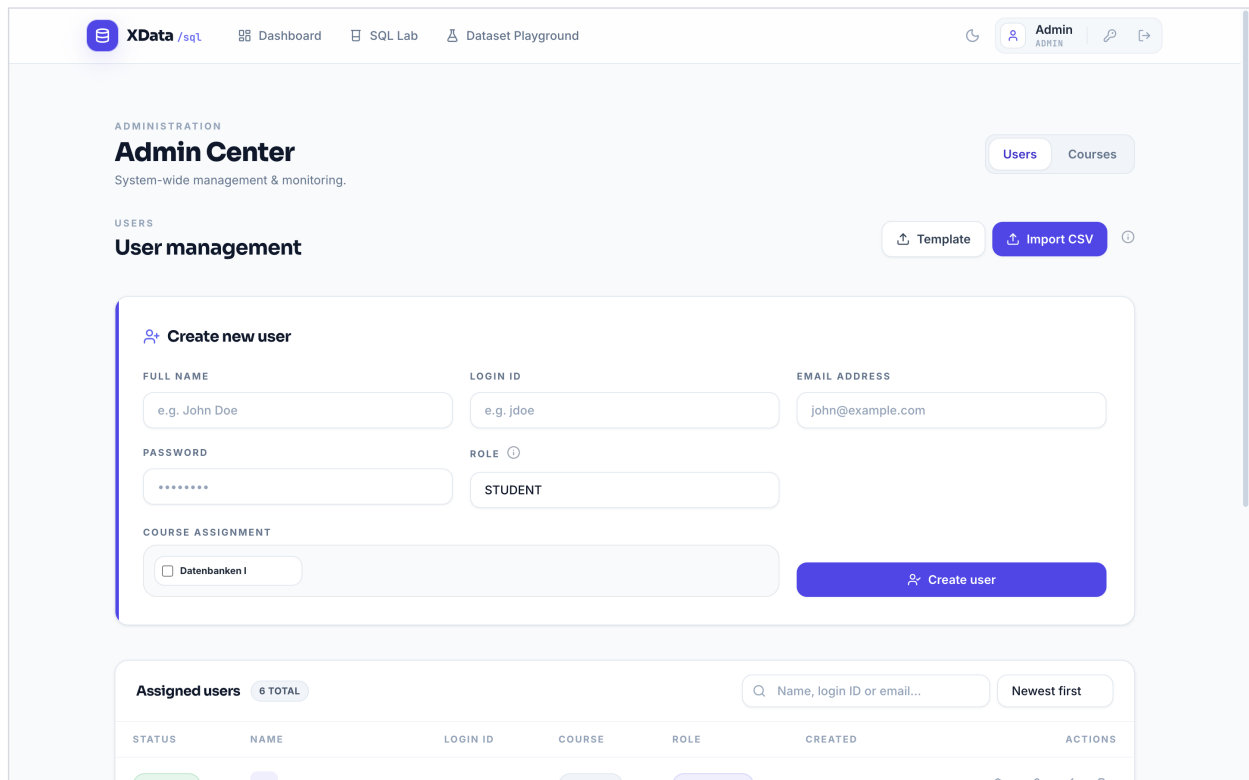


Figure 25: The interface in light mode (toggled from the top bar).

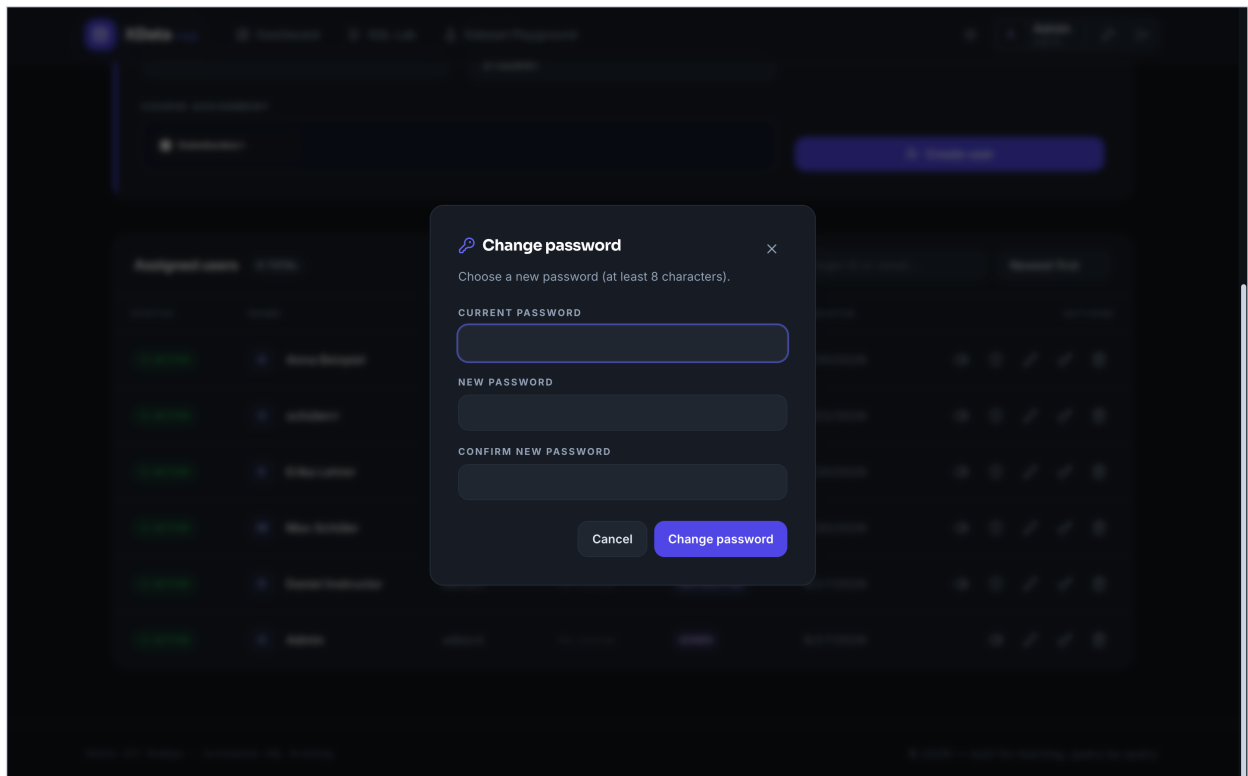


Figure 26: The change-password dialog (key icon, top bar).

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